Lecture 6 Options

1. Introduction to Options

An **option** is a financial contract between a buyer and a seller that gives the buyer the right, but not the obligation, to buy or sell an underlying asset at a specified price on or before a specified date.

In this chapter, we focus on **European options**, which can only be exercised at maturity.

2. Call and Put Options

Call Option(看涨期权)

A **call option** gives the buyer the right to purchase the underlying asset at a predetermined price, called the **strike price** (K), on the expiration date.

Put Option（看跌期权）

A **put option** gives the buyer the right to sell the underlying asset at the strike price (K) on the expiration date.

3. Payoff Functions

Call Option Payoff

At expiration time t, the payoff of a call option is:

$$\text{Payoff} = \max(S(t) - K, 0) = (S(t) - K)^+$$

Put Option Payoff

At expiration time t, the payoff of a put option is:

$$\text{Payoff} = \max(K - S(t), 0) = (K - S(t))^+$$

4. Basic Option Terminology

- **Strike Price（执行价）** (K): the agreed-upon price for the underlying asset.
- **Spot Price(股票价格)** (S): current price of the underlying stock.
- **Expiration Date（到期日）**: date when the option contract expires.
- **Time to Expiration（到期期限）** (t): time remaining until expiration.

$$t = T - t_{\text{now}}$$

- **Premium(期权费)**: the price paid by the buyer to acquire the option contract.
- **In-the-Money (ITM)**:
  - 期权当前有内在价值，行权有利可图。
  - Call: $S > K$

  - Put: $S < K$

- **Out-of-the-Money (OTM)**:
  - 期权当前没有内在价值，不适合行权。
  - Call: $S < K$
  - Put: $S > K$

- Call = 未来我有权低价买入，所以我希望未来价格上涨。
- Put = 未来我有权高价卖出，所以我希望未来价格下跌。
- 无论 Call 还是 Put，最大亏损都不会超过买入合约时支付的 premium；
- 期权是一种典型的 非对称收益工具，适合用来做风险对冲或投机操作。

5. Profit and Loss (P&L) Diagrams

Call Option Buyer

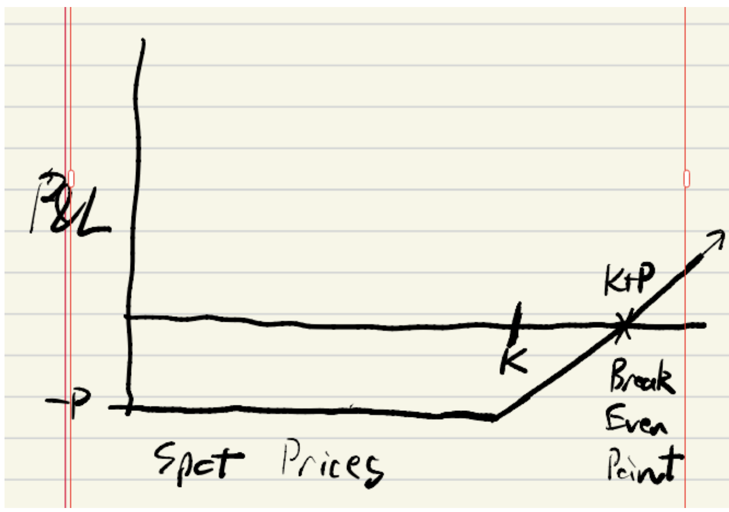
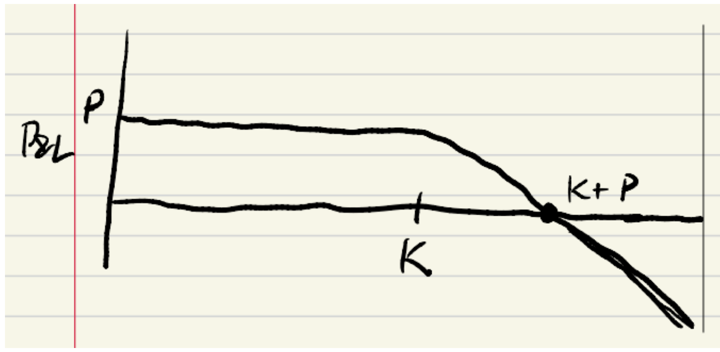


Abbildung 1: Profit and Loss Diagram of a Call Option Buyer

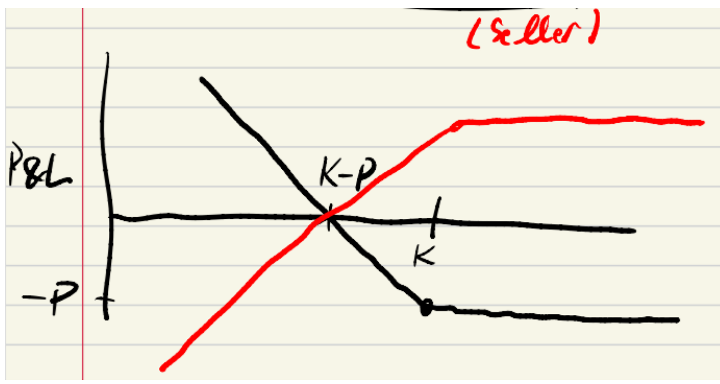
$$\text{Profit} = \begin{cases} S - K - P, & \text{if } S \geq K \text{行权获利} \\ -P, & \text{if } S < K \text{期权不行权，损失为已支付的 premium} \end{cases}$$

Break-even point: $S = K + P$.

Call Option Seller



$$\text{Profit} = \begin{cases} P, & \text{if } S \le K \\ P - (S-K), & \text{if } S > K \end{cases}$$



Put Option Buyer

$$\text{Profit} = \begin{cases} K - S - P, & \text{if } S \le K \\ -P, & \text{if } S > K \end{cases}$$

Put Option Seller

$$\text{Profit} = \begin{cases} P, & \text{if } S \ge K \\ P - (K-S), & \text{if } S < K \end{cases}$$

6. Goal: Pricing Options

We want to compute the fair price of a European call/put option today.

$$C(0) = \mathbb{E}\left[(S(t) - K)^+\right]$$

当前价格等于未来所有可能收益的加权平均值，这里的加权就是“发生该情况的概率”。

$$P(0) = \mathbb{E}\left[(K - S(t))^+\right]$$

7. Modeling Stock Prices

We assume the stock price $S(t)$ follows **Geometric Brownian Motion (GBM)**:

$$S(t) = S(0)e^{\left(-\frac{\sigma^2}{2}t+\sigma\sqrt{t}Z\right)}$$

where:

- $Z \sim \mathcal{N}(0,1)$ (standard normal distribution)
- σ is the volatility (standard deviation of log-returns)

8. The Black-Scholes Formula（期权定价公式）

Call Option Price (看涨期权价格)

$$C(0) = S(0)\Phi(d_1) - K\Phi(d_2)$$

- $\Phi(d_1), \Phi(d_2)$：标准正态分布的累积分布函数（CDF），表示“涨上去的概率”。
- Call 买方的价值 = 股票有可能涨上去 × 涨上去时的收益 − 未来支付行权价的期望值。

Put Option Price (看跌期权价格)

$$P(0) = K\Phi(-d_2) - S(0)\Phi(-d_1)$$

Put 买方的价值 = 股票跌破行权价时的期望收益
where:

$$d_1 = \frac{\ln\left(\frac{S(0)}{K}\right) + \frac{\sigma^2}{2}t}{\sigma\sqrt{t}}, \quad d_2 = d_1 - \sigma\sqrt{t}$$

and $\Phi(\cdot)$ is the standard normal cumulative distribution function:

$$\Phi(y) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{y} e^{-\frac{z^2}{2}}dz$$

- 数学上就是标准正态分布的累计概率函数（累积分布函数）
- 例如 $\Phi(1.96) \approx 0.975$：意味着正态分布下 97.5% 的概率落在这个点左侧

例子:定价一个看涨期权

- 当前股价：$S(0) = 100$
- 行权价：$K = 105$
- 波动率：$\sigma = 0.2$
- 到期时间：$t = 1$ 年
- 无风险利率暂时忽略（即不贴现）

1. 计算 d_1, d_2（衡量“股票价格超过行权价的可能性”的指标）：

$$d_1 = \frac{\ln(100/105) + \frac{0.2^2}{2} \cdot 1}{0.2 \cdot \sqrt{1}} = \frac{-0.0488 + 0.02}{0.2} = \frac{-0.0288}{0.2} \approx -0.144$$

$$d_2 = d_1 - \sigma\sqrt{t} = -0.144 - 0.2 = -0.344$$

2. 查表或计算： 通过“正态分布表”（Z-Table）查值。Z 表给出了：从负无穷到某个 z 值的累积概率 $\Phi(z)$

$$\Phi(d_1) \approx 0.443, \quad \Phi(d_2) \approx 0.366$$

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[2]: from scipy.stats import norm
```

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[4]: #  d1 和 d2 的数值
d1 = -0.1439
d2 = -0.344
```

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[6]: # 计算标准正态分布的CDF值
phi_d1 = norm.cdf(d1)
phi_d2 = norm.cdf(d2)
```

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[8]: print(f"Phi(d1) = {phi_d1:.3f}")
print(f"Phi(d2) = {phi_d2:.3f}")

Phi(d1) = 0.443
Phi(d2) = 0.365
```

3. 套用公式：

$$C(0) = S(0) \cdot \Phi(d_1) - K \cdot \Phi(d_2)$$

$$C(0) = 100 \cdot 0.443 - 105 \cdot 0.366 \approx 44.3 - 38.4 = 5.9$$

✓ **所以这个期权现在值 \$5.90。**

9. Sketch of Black-Scholes Derivation

使用 Indicator 函数拆分期望，We use:

$$\mathbb{E}\left[(S(t) - K)^+\right] = \mathbb{E}\left[(S(t) - K)1_{S(t) \geq K}\right]$$

- 因为 $(S(t) - K)^+$ 其实就是当 $S(t) \geq K$ 时才有值，否则为 0;
- 所以我们可以用 Indicator 函数 $\mathbf{1}_{\{S(t) \geq K\}}$ 把它表达出来。

This can be rewritten as（拆成两个期望）：

$$\mathbb{E}\left[(S(t) - K)^+\right] = \mathbb{E}\left[S(t) \cdot 1_{S(t) \geq K}\right] - K\mathbb{E}\left[1_{S(t) \geq K}\right]$$

- $S(t) \cdot 1$: 只在“股票涨过 K”时考虑股票的价格;
- $K \cdot 1$: 只在“涨过 K”时才减去行权价。

The two expectations are computed separately, yielding:

$$\mathbb{E}\left[S(t)\cdot 1_{S(t)\geq K}\right] = S(0)\Phi(d_1)$$
$$\mathbb{E}\left[1_{S(t)\geq K}\right] = \Phi(d_2)$$

1. **第一个期望**:
$$\mathbb{E}\left[S(t)\cdot \mathbf{1}_{\{S(t)\geq K\}}\right] = S(0)\Phi(d_1)$$
   - 这是未来股票价格超过 K 时的加权平均值
   - 实际上利用了股票价格服从几何布朗运动，再转为正态分布后积分得出（涉及风险中性概率测度）

2. **第二个期望**:
$$\mathbb{E}\left[\mathbf{1}_{\{S(t)\geq K\}}\right] = \Phi(d_2)$$
   - 这是股票价格涨过 K 的 "概率"
   - 就是 $Z >$ 某个数 的概率，直接查标准正态表得到

Thus,

$$C(0) = S(0)\Phi(d_1) - K\Phi(d_2)$$

Problem Setup

We assume that the stock price $S(t)$ at expiration time t follows a Geometric Brownian Motion (GBM):

$$S(t) = S(0)e^{-\frac{\sigma^2}{2}t+\sigma\sqrt{t}Z},$$

where $Z \sim \mathcal{N}(0,1)$ is a standard normal random variable, and σ is the volatility.

We want to compute the value of the European call option:

$$C(0) = \mathbb{E}\left[(S(t)-K)^+\right],$$

where K is the strike price and $(x)^+ = \max(x,0)$.

Splitting the Expectation

We rewrite the expectation as:

$$\mathbb{E}\left[(S(t)-K)^+\right] = \mathbb{E}\left[(S(t)-K)\cdot 1_{\{S(t)\geq K\}}\right].$$

Using linearity of expectation:

$$\mathbb{E}\left[(S(t)-K)^+\right] = \mathbb{E}\left[S(t)\cdot 1_{\{S(t)\geq K\}}\right] - K\mathbb{E}\left[1_{\{S(t)\geq K\}}\right].$$

Thus we need to compute two parts:

- $\mathbb{E}\left[1_{\{S(t)\geq K\}}\right]$
- $\mathbb{E}\left[S(t)\cdot 1_{\{S(t)\geq K\}}\right]$

Computing $\mathbb{E}\left[1_{\{S(t)\geq K\}}\right]$

We start by solving for the condition $S(t) \geq K$:

$$S(0)e^{-\frac{\sigma^2}{2}t+\sigma\sqrt{t}Z} \geq K.$$

Taking logarithms:

$$-\frac{\sigma^2}{2}t + \sigma\sqrt{t}Z \geq \ln\left(\frac{K}{S(0)}\right).$$

Rearranging:

$$Z \geq \frac{\ln\left(\frac{K}{S(0)}\right) + \frac{\sigma^2}{2}t}{\sigma\sqrt{t}}.$$

Define:

$$d_2 = \frac{\ln\left(\frac{S(0)}{K}\right) - \frac{\sigma^2}{2}t}{\sigma\sqrt{t}}.$$

Thus:

$$\mathbb{P}(S(t) \geq K) = \mathbb{P}(Z \geq d_2) = 1 - \Phi(d_2).$$

So:

$$\mathbb{E}\left[1_{\{S(t)\geq K\}}\right] = \Phi(d_2).$$

Computing $\mathbb{E}\left[S(t) \cdot 1_{\{S(t)\geq K\}}\right]$

We rewrite:

$$\mathbb{E}\left[S(t) \cdot 1_{\{S(t)\geq K\}}\right] = S(0)\mathbb{E}\left[e^{-\frac{\sigma^2}{2}t+\sigma\sqrt{t}Z} \cdot 1_{\{Z\geq d_2\}}\right].$$

Factor out $S(0)$:

$$= S(0)e^{-\frac{\sigma^2}{2}t}\mathbb{E}\left[e^{\sigma\sqrt{t}Z} \cdot 1_{\{Z\geq d_2\}}\right].$$

Note that $e^{\sigma\sqrt{t}Z}$ is log-normal, and rewrite:

$$e^{\sigma\sqrt{t}Z} = e^{\frac{\sigma^2 t}{2}} \cdot e^{\sigma\sqrt{t}(Z-\sigma\sqrt{t}/2)}.$$

Thus,

$$\mathbb{E}\left[S(t) \cdot 1_{\{S(t)\geq K\}}\right] = S(0)\mathbb{E}\left[e^{\sigma\sqrt{t}Z} \cdot 1_{\{Z\geq d_2\}}\right] = S(0)\Phi(d_1),$$

where

$$d_1 = \frac{\ln\left(\frac{S(0)}{K}\right) + \frac{\sigma^2}{2}t}{\sigma\sqrt{t}} = d_2 + \sigma\sqrt{t}.$$

Conclusion

Now putting both pieces together:

$$C(0) = S(0)\Phi(d_1) - K\Phi(d_2).$$

Put Option Formula (Put-Call Parity)

The formula for the European put option can be derived using put-call parity:

$$C(0) - P(0) = S(0) - K.$$

Thus:

$$P(0) = C(0) + K - S(0) = -S(0)\Phi(-d_1) + K\Phi(-d_2).$$

Summary of Black-Scholes Formula

$$C(0) = S(0)\Phi(d_1) - K\Phi(d_2),$$
$$P(0) = K\Phi(-d_2) - S(0)\Phi(-d_1),$$

with

$$d_1 = \frac{\ln\left(\frac{S(0)}{K}\right) + \frac{\sigma^2}{2}t}{\sigma\sqrt{t}}, \quad d_2 = d_1 - \sigma\sqrt{t}.$$

Standard Normal CDF

$$\Phi(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{y} e^{-z^2/2}\, dz.$$

10. Call-Put Parity

The relationship between call and put prices is:

$$C(0) - P(0) = S(0) - K$$

This allows us to compute one option price from the other.

11. Remarks

- Option prices incorporate expectations of future volatility and market behavior.
- Selling options carries large risk if not managed properly.
- The Black-Scholes model assumes:
  - No arbitrage
  - Constant interest rate (we assumed $r = 0$ here)
  - Constant volatility
  - Log-normal stock price distribution